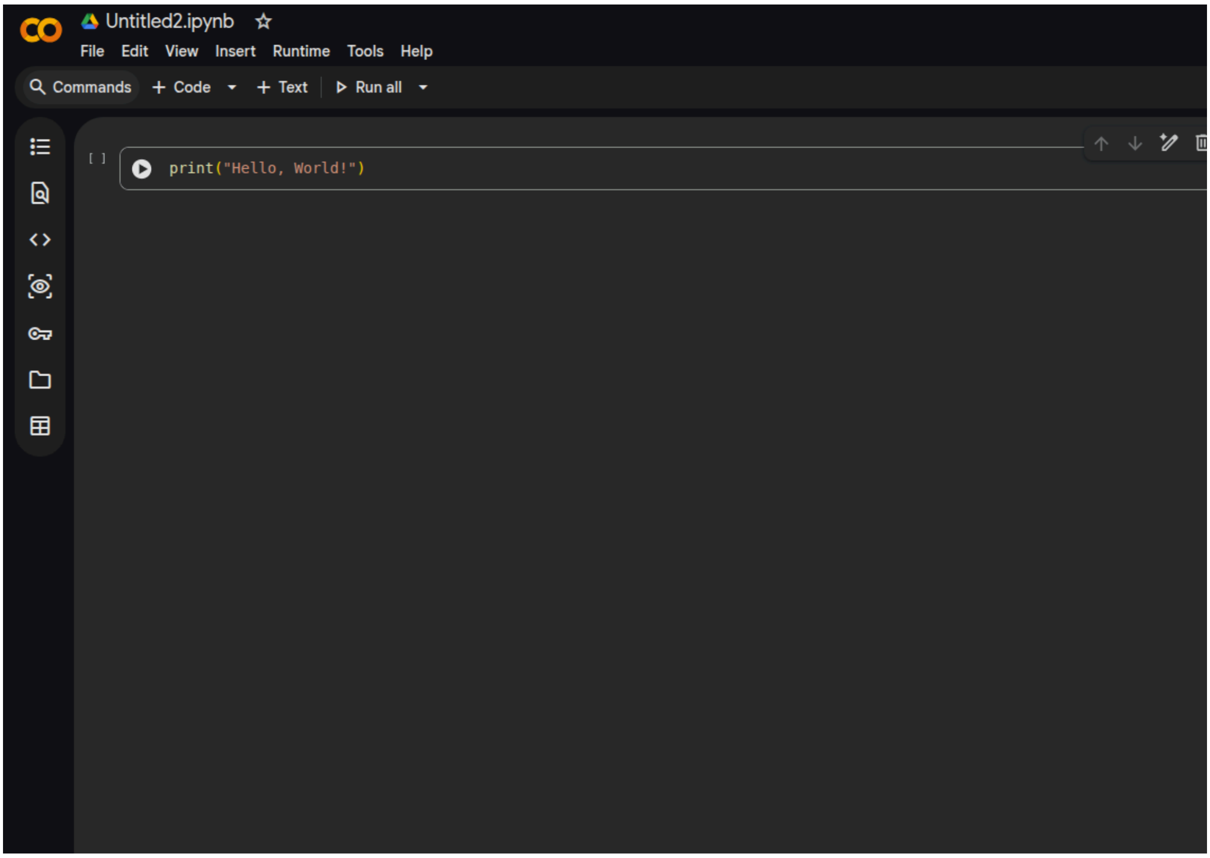
Python for Beginners

Setting up your environment

1. Go to Google Colab: https://colab.research.google.com/
    a. You will need a google account to sign in!
2. On the top left of the screen, click “File”, then “New Notebook in Drive”
3. You should see a screen that looks something like this below:



4. This is where you will type your code!
    a. The white button will run your code, but before we do that we need some code to write!

Printing

We can now type some code and run it - if you have problems, that’s ok. Put up your hand and ask an educator for help - that’s our job!

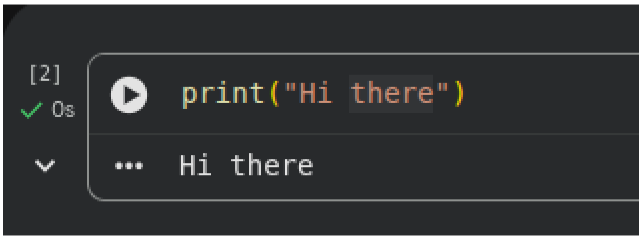
The first line of code almost every programmer writes is:

```
print(“Hello, World!”)
```

Copy and paste the above line of code into Google Colab. Click the white button to run your code, and you should see the code printing out “Hello, World!” like you see below:

This line of code simply tells the computer: *Hey, can you **print** the words inside the quotation (“) marks?*

You can print out whatever you like. For example, change the words inside the quotation marks to something else. For example:



Play around with this yourself! Change the words to whatever you want. The best way to learn programming is to mess around and tinker with things - that’s where the real understanding happens!

Before we move on, let’s do some quick exercises:

1. Print out the name of your favourite movie to the screen (anything but Iron Man 3 ok)
2. Print out the name of your favourite book or TV show to the screen
3. Print out the name of your favourite video game to the screen - if you don’t have one, print out the name of your favourite hobby to the screen, e.g. crocheting

Before you move on, put your hand up and ask an educator if you think you solved these exercises.

Variables

You were able to print out some stuff to the screen. If you had some issues, raise your hand and ask an educator for help, we’ll be happy to help. If you were ok with that, we can move on now to something different.

We will not talk about a programming concept called **variables.** In Python, a *variable* is something that stores a piece of information - a phone number, a name, address, anything!

Let’s create a variable in python so you can get a better idea of what this means. Look at the line of code below:

my_name = “Brendan”

The line of code above creates a *variable* called `my_name`, and a piece of information is stored inside the variable. In this case, my name is stored as a piece of information in the variable.

To get a better understanding of how variables work, copy/paste the two lines of code below into Google Colab:

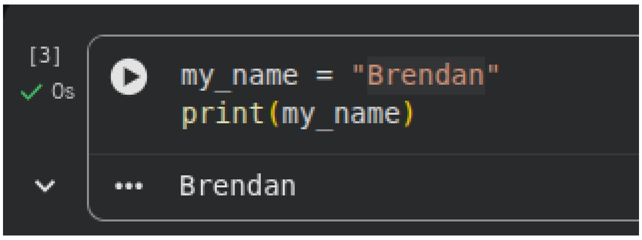
```
my_name = “Brendan”
print(my_name)
```

Before you run the code, what do you think the output of this program will be?

1. `Brendan`
2. `my_name`

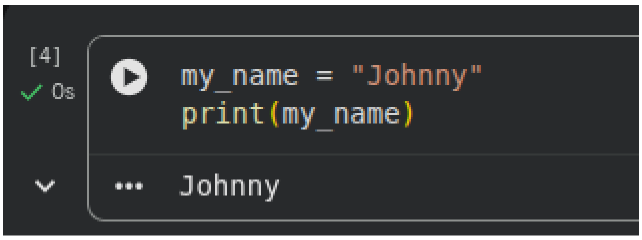
Take a guess - it’s ok if you’re wrong because you’ll find out why later.

When you run the code, this is the output you will see:



The program printed out the **information** that the variable contains (Brendan), not the name of the variable (`my_name`). This might seem a bit confusing, but that’s ok. Continue on, and it will make more sense

Now, change the information of the variable. For example, something like:



Can you notice a pattern now? Variables **store information**. In this case, we created a variable called “`my_name`”, and we stored my name as information in this variable.

Don’t just move on to the next exercise right away. Tinker with the code yourself - change the information that’s stored in the variable - for example, change the name to your own name! Or your favourite movie, it’s up to you.

Now, create a variable called `my_age` and store your age in this variable. Then, print out your age. The answer for this is on the next page, but try this out yourself before checking the right answer!

Solution:

```
[6]
0s
my_age = "23"
print(my_age)

... 23
```

Exercises:

1. Create a variable called my_favourite_movie, and store the name of your favourite movie inside this variable. Print it out to the screen.
2. Create a variable called my_favourite_hobby, and store the name of your favourite hobby inside this variable. Print it out to the screen.
3. Create a variable called my_favourite_book, and store the name of your favourite book inside this variable. Print it out to the screen.

Before you move on, put your hand up and ask an educator if you think you solved these exercises.

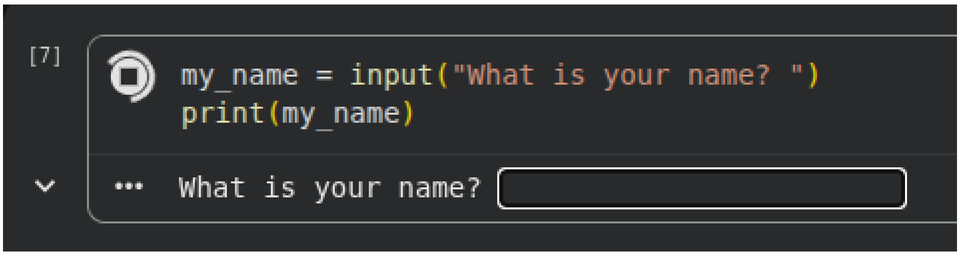
User Input

Ok, we've shown you how to print to the screen, and to store different information in variables. This stuff is really important for programming, and it is the building blocks of everything you will do in programming. If there is anything you don't understand, that's normal! These things can be a bit confusing at first, and they took me a while to get my head around them. Ask an educator for help with anything you've struggled to understand so far.

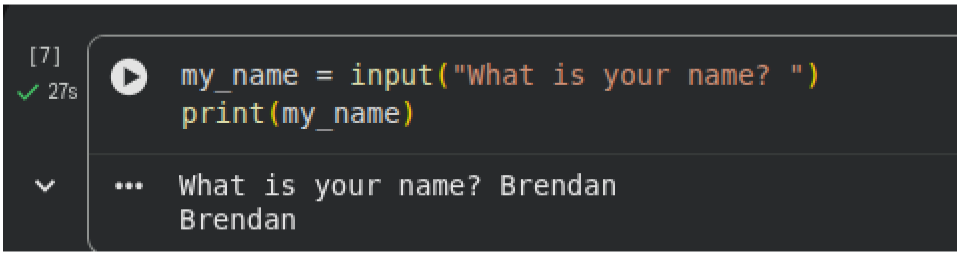
Now, let's take some input from the user. Copy the following line of code below:

```
my_name = input("What is your name? ")
print(my_name)
```

The code above will ask the user for their name, the user will type in their name, and then it will print it out to the screen. This might not make sense, so run the code yourself and you will understand it much better!



When you first run the code, you'll see a text box - this is asking you to type in something!



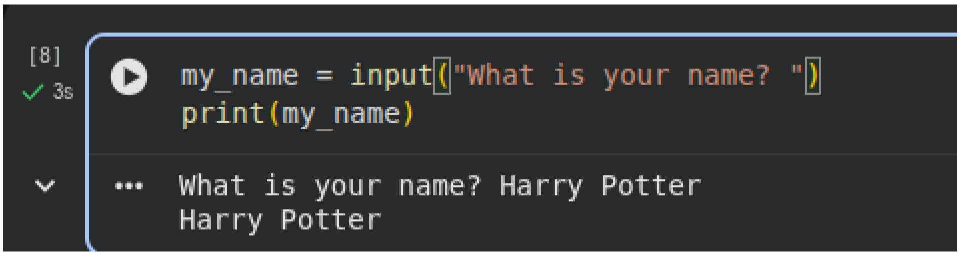
After typing in "Brendan", the code printed out my name.

To understand how this worked, let's look at the code again:

```
my_name = input(“What is your name? “)
print(my_name)
```

`input()` asks the user to type in something. When the user types in something, we will store this information in a variable called `my_name`, and then we will print out `my_name`. It's ok if you don't understand this, I didn't get it right away either. It will make more sense as you code more.

Run the code, and type in whatever you like - it doesn't have to be your name! It could be your age - anything! for example:



When the computer program asks me to type something into the box, it will print that out to the screen. Run this code a few times, and type in a bunch of different stuff: your favourite movie, book, character, anything!

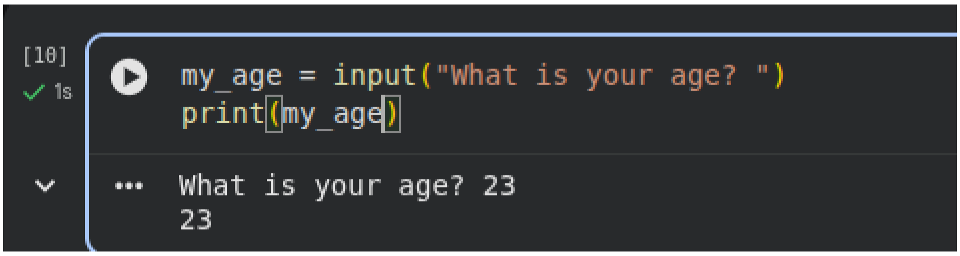
Ok, you should have run the code a few times. Now, let's change the question we are asking the user. Instead, let's ask the user for their age. We can do this by using the two lines of code below:

```
my_age = input("What is your age? ")
print(my_age)
```

Did you notice what we changed above? Take a close look, it will help you understand:

1. Instead of asking the user “What is your name? “, we asked “What is your age? “
2. Because we are asking for the user’s age and not their name, we changed the name of the variable from `my_name` to `my_age`

Now, run the code, you should get something like:



Run the code a few times, with different numbers, just to try it out. It will help your understanding.

Exercises:

1. Ask the user “What is your favourite movie? “, and print the result to the screen.
2. Ask the user “What is your favourite book? “, and print the result to the screen.
3. Ask the user “What is your favourite hobby? “, and print the result to the screen.

Before you move on, put your hand up and ask an educator if you think you solved these exercises.